

Polynomial End Behavior, Zeros & Multiplicity

ANSWER KEY



End behavior of polynomial functions

- 1 State the end behavior of $f(x) = -2x^5 + 3x^2 - 1$, using the leading term.

Leading term $-2x^5$: odd degree, negative leading coefficient. As $x \rightarrow -\infty$, $f(x) \rightarrow +\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow -\infty$.

- 2 State the end behavior of $f(x) = 3x^4 - 5x^3 + 2$.

Leading term $3x^4$: even degree, positive leading coefficient. As $x \rightarrow \pm\infty$, $f(x) \rightarrow +\infty$.

Zeros and their multiplicities

- 3 Find the zeros of $f(x) = (x - 2)^3(x + 1)^2$ and state the multiplicity and graphical behavior at each.

$x = 2$ has multiplicity 3 (odd): the graph crosses the axis, flattening near $x = 2$. $x = -1$ has multiplicity 2 (even): the graph touches the axis and turns back (a local extremum), not crossing.

- 4 The graph shows $f(x) = (x + 2)(x - 1)^2$, illustrating a crossing zero at $x = -2$ and a touching zero at $x = 1$.

At $x = -2$ (multiplicity 1, odd): the graph crosses the x-axis. At $x = 1$ (multiplicity 2, even): the graph touches the axis and turns back without crossing, consistent with the flattened touch point shown.

The Rational Zero Theorem

- 5 List all possible rational zeros of $f(x) = 2x^3 - 3x^2 - 8x + 12$ using the Rational Zero Theorem, then find the actual zeros.

Possible rational zeros: $\pm 1, \pm 2, \pm 3, \pm 4, \pm 6, \pm 12, \pm \frac{1}{2}, \pm \frac{3}{2}$ (factors of 12 over factors of 2). Testing $x = 2$: $2(8) - 3(4) - 16 + 12 = 16 - 12 - 16 + 12 = 0$ ✓. Dividing:
 $2x^3 - 3x^2 - 8x + 12 = (x - 2)(2x^2 + x - 6) = (x - 2)(2x - 3)(x + 2)$. Zeros: $x = 2, \frac{3}{2}, -2$.

Constructing a polynomial from its zeros

- 6 Write a polynomial function of least degree with real coefficients, zeros at $x = -1$ (multiplicity 2) and $x = 3$ (multiplicity 1), and a leading coefficient of 1.

$$f(x) = (x + 1)^2(x - 3) = (x^2 + 2x + 1)(x - 3) = x^3 - x^2 - 5x - 3.$$

- 7 A polynomial has zeros at $x = 0$, $x = 2$, and $x = -3$, all with multiplicity 1, and passes through $(1, 8)$. Find the polynomial in the form $f(x) = a \cdot x(x - 2)(x + 3)$.

$$f(1) = a(1)(1 - 2)(1 + 3) = a(1)(-1)(4) = -4a = 8, \text{ so } a = -2. f(x) = -2x(x - 2)(x + 3).$$

Local extrema and turning points

- 8 A polynomial of degree n has at most how many turning points (local extrema)? State the rule and apply it to a degree-5 polynomial.

A degree- n polynomial has at most $n - 1$ turning points. A degree-5 polynomial has at most 4 turning points.

- 9 The graph of a polynomial shows exactly 3 turning points. What is the minimum possible degree of the polynomial?

Since turning points $\leq n - 1$, having 3 turning points requires $n - 1 \geq 3$, so $n \geq 4$. The minimum degree is 4.

Average rate of change and concavity

- 10 Find the average rate of change (AROC) of $f(x) = x^3 - 4x$ on $[1, 3]$.

$$\text{AROC} = \frac{f(3) - f(1)}{3 - 1} = \frac{(27 - 12) - (1 - 4)}{2} = \frac{15 - (-3)}{2} = \frac{18}{2} = 9.$$

- 11 For $f(x) = x^3 - 4x$, compare the AROC on $[0, 1]$ and $[1, 2]$, and use the comparison to determine whether f is concave up or concave down on this interval.

AROC on $[0, 1]$: $\frac{f(1) - f(0)}{1 - 0} = \frac{-3 - 0}{1} = -3$. AROC on $[1, 2]$: $\frac{f(2) - f(1)}{2 - 1} = \frac{0 - (-3)}{1} = 3$. Since the AROC is increasing from -3 to 3 , f is concave up on $[0, 2]$.

Sign charts and interval analysis

- 12 Build a sign chart for $f(x) = (x + 1)(x - 2)(x - 4)$ and state the intervals where $f(x) > 0$.

Zeros at $x = -1, 2, 4$ split the line into four intervals. Testing a point in each: $x < -1$ gives $f < 0$; $-1 < x < 2$ gives $f > 0$; $2 < x < 4$ gives $f < 0$; $x > 4$ gives $f > 0$. So $f(x) > 0$ on $(-1, 2) \cup (4, \infty)$.

- 13 Use a sign chart to solve the inequality $(x - 1)^2(x + 3) \leq 0$.

Zeros at $x = 1$ (multiplicity 2, sign does not change) and $x = -3$ (multiplicity 1, sign changes). Testing intervals: for $x < -3$, expression is negative; for $-3 < x < 1$, positive; for $x > 1$, positive. So the expression is ≤ 0 on $(-\infty, -3] \cup \{1\}$.

Visualizing end behavior families

14 The graph shows $f(x) = x^4 - 5x^2 + 4$, a degree-4 polynomial with four real zeros.

a State the end behavior of the graph as $x \rightarrow \pm \infty$. Even degree, positive leading coefficient: $x \rightarrow \pm \infty$, $f(x) \rightarrow + \infty$, matching the upward arms shown on both sides.

b Identify the four zeros shown and state whether each is a crossing or touching zero. Zeros at $x = -2, -1, 1, 2$, each multiplicity 1 (odd), so all four are crossing zeros.

c How many turning points does the graph display, and is this consistent with the degree? The graph shows 3 turning points, which is exactly $n - 1 = 4 - 1 = 3$ for a degree-4 polynomial — consistent with the maximum allowed.

Putting it together: a full analysis

15 For $f(x) = x^3 + 2x^2 - 5x - 6$: verify that $x = -1$ is a zero, factor completely, then describe the end behavior and the behavior at each zero.

$f(-1) = -1 + 2 + 5 - 6 = 0$ ✓. Dividing by $(x + 1)$:

$x^3 + 2x^2 - 5x - 6 = (x + 1)(x^2 + x - 6) = (x + 1)(x + 3)(x - 2)$. Zeros at $x = -1, -3, 2$, all multiplicity 1 (crossing). End behavior: odd degree, positive leading coefficient, so $f(x) \rightarrow -\infty$ as $x \rightarrow -\infty$ and $f(x) \rightarrow +\infty$ as $x \rightarrow +\infty$.

Multiple choice — easy 15

16 State the end behavior of a polynomial with leading term x^7 .

a. As $x \rightarrow -\infty$, $f(x) \rightarrow +\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow -\infty$

b. As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow +\infty$ ✓

c. As $x \rightarrow \pm \infty$, $f(x) \rightarrow -\infty$

d. As $x \rightarrow \pm \infty$, $f(x) \rightarrow +\infty$

The leading term x^7 has odd degree and a positive coefficient. As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow +\infty$.

17 State the end behavior of a polynomial with leading term $9x^5$.

- a. As $x \rightarrow -\infty$, $f(x) \rightarrow +\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow -\infty$
- b. As $x \rightarrow \pm\infty$, $f(x) \rightarrow -\infty$
- c. As $x \rightarrow \pm\infty$, $f(x) \rightarrow +\infty$
- d. As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow +\infty$ ✓

The leading term $9x^5$ has odd degree and a positive coefficient. As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow +\infty$.

18 State the end behavior of a polynomial with leading term $-5x^6$.

- a. As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow +\infty$
- b. As $x \rightarrow \pm\infty$, $f(x) \rightarrow +\infty$
- c. As $x \rightarrow \pm\infty$, $f(x) \rightarrow -\infty$ ✓
- d. As $x \rightarrow -\infty$, $f(x) \rightarrow +\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow -\infty$

The leading term $-5x^6$ has even degree and a negative coefficient. As $x \rightarrow \pm\infty$, $f(x) \rightarrow -\infty$.

19 State the end behavior of a polynomial with leading term $8x^2$.

- a. As $x \rightarrow \pm\infty$, $f(x) \rightarrow +\infty$ ✓
- b. As $x \rightarrow -\infty$, $f(x) \rightarrow +\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow -\infty$
- c. As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow +\infty$
- d. As $x \rightarrow \pm\infty$, $f(x) \rightarrow -\infty$

The leading term $8x^2$ has even degree and a positive coefficient. As $x \rightarrow \pm\infty$, $f(x) \rightarrow +\infty$.

20 State the end behavior of a polynomial with leading term $6x^2$.

- a. As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow +\infty$
- b. As $x \rightarrow -\infty$, $f(x) \rightarrow +\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow -\infty$
- c. As $x \rightarrow \pm\infty$, $f(x) \rightarrow -\infty$
- d. As $x \rightarrow \pm\infty$, $f(x) \rightarrow +\infty$ ✓

The leading term $6x^2$ has even degree and a positive coefficient. As $x \rightarrow \pm\infty$, $f(x) \rightarrow +\infty$.

21 State the end behavior of a polynomial with leading term $2x^7$.

- a. As $x \rightarrow -\infty$, $f(x) \rightarrow +\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow -\infty$
- b. As $x \rightarrow \pm\infty$, $f(x) \rightarrow +\infty$
- c. As $x \rightarrow \pm\infty$, $f(x) \rightarrow -\infty$
- d. As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow +\infty$ ✓

The leading term $2x^7$ has odd degree and a positive coefficient. As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow +\infty$.

22 State the end behavior of a polynomial with leading term x^6 .

- a. As $x \rightarrow \pm\infty$, $f(x) \rightarrow +\infty$ ✓
- b. As $x \rightarrow \pm\infty$, $f(x) \rightarrow -\infty$
- c. As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow +\infty$
- d. As $x \rightarrow -\infty$, $f(x) \rightarrow +\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow -\infty$

The leading term x^6 has even degree and a positive coefficient. As $x \rightarrow \pm\infty$, $f(x) \rightarrow +\infty$.

23 State the end behavior of a polynomial with leading term $4x^3$.

- a. As $x \rightarrow \pm \infty$, $f(x) \rightarrow -\infty$
- b. As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow +\infty$ ✓
- c. As $x \rightarrow -\infty$, $f(x) \rightarrow +\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow -\infty$
- d. As $x \rightarrow \pm \infty$, $f(x) \rightarrow +\infty$

The leading term $4x^3$ has odd degree and a positive coefficient. As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow +\infty$.

24 State the end behavior of a polynomial with leading term $-x^7$.

- a. As $x \rightarrow \pm \infty$, $f(x) \rightarrow +\infty$
- b. As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow +\infty$
- c. As $x \rightarrow -\infty$, $f(x) \rightarrow +\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow -\infty$ ✓
- d. As $x \rightarrow \pm \infty$, $f(x) \rightarrow -\infty$

The leading term $-x^7$ has odd degree and a negative coefficient. As $x \rightarrow -\infty$, $f(x) \rightarrow +\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow -\infty$.

25 State the end behavior of a polynomial with leading term $-2x^7$.

- a. As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow +\infty$
- b. As $x \rightarrow \pm \infty$, $f(x) \rightarrow -\infty$
- c. As $x \rightarrow -\infty$, $f(x) \rightarrow +\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow -\infty$ ✓
- d. As $x \rightarrow \pm \infty$, $f(x) \rightarrow +\infty$

The leading term $-2x^7$ has odd degree and a negative coefficient. As $x \rightarrow -\infty$, $f(x) \rightarrow +\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow -\infty$.

26 State the end behavior of a polynomial with leading term $6x^5$.

- a. As $x \rightarrow \pm \infty$, $f(x) \rightarrow +\infty$
- b. As $x \rightarrow \pm \infty$, $f(x) \rightarrow -\infty$
- c. As $x \rightarrow -\infty$, $f(x) \rightarrow +\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow -\infty$
- d. As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow +\infty$ ✓

The leading term $6x^5$ has odd degree and a positive coefficient. As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow +\infty$.

27 State the end behavior of a polynomial with leading term $4x^2$.

- a. As $x \rightarrow -\infty$, $f(x) \rightarrow +\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow -\infty$
- b. As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow +\infty$
- c. As $x \rightarrow \pm \infty$, $f(x) \rightarrow +\infty$ ✓
- d. As $x \rightarrow \pm \infty$, $f(x) \rightarrow -\infty$

The leading term $4x^2$ has even degree and a positive coefficient. As $x \rightarrow \pm \infty$, $f(x) \rightarrow +\infty$.

28 State the end behavior of a polynomial with leading term $8x^5$.

- a. As $x \rightarrow -\infty$, $f(x) \rightarrow +\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow -\infty$
- b. As $x \rightarrow \pm \infty$, $f(x) \rightarrow +\infty$
- c. As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow +\infty$ ✓
- d. As $x \rightarrow \pm \infty$, $f(x) \rightarrow -\infty$

The leading term $8x^5$ has odd degree and a positive coefficient. As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow +\infty$.

29 State the end behavior of a polynomial with leading term $7x^7$.

- a. As $x \rightarrow -\infty$, $f(x) \rightarrow +\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow -\infty$
- b. As $x \rightarrow \pm\infty$, $f(x) \rightarrow +\infty$
- c. As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow +\infty$ ✓
- d. As $x \rightarrow \pm\infty$, $f(x) \rightarrow -\infty$

The leading term $7x^7$ has odd degree and a positive coefficient. As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow +\infty$.

30 State the end behavior of a polynomial with leading term $-7x^2$.

- a. As $x \rightarrow \pm\infty$, $f(x) \rightarrow -\infty$ ✓
- b. As $x \rightarrow -\infty$, $f(x) \rightarrow +\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow -\infty$
- c. As $x \rightarrow \pm\infty$, $f(x) \rightarrow +\infty$
- d. As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow +\infty$

The leading term $-7x^2$ has even degree and a negative coefficient. As $x \rightarrow \pm\infty$, $f(x) \rightarrow -\infty$.

Multiple choice — medium 15

31 Find the degree of $f(x) = (x - 4)^1(x + 6)^2(x + 7)^1$.

- a. 2
- b. 5
- c. 7
- d. 4 ✓

The degree is the sum of the multiplicities: $1 + 2 + 1 = 4$.

32 Find the degree of $f(x) = (x + 2)^2(x + 3)^1(x + 8)^1$.

- a. 5
- b. 2
- c. 6
- d. 4 ✓

The degree is the sum of the multiplicities: $2 + 1 + 1 = 4$.

33 Find the degree of $f(x) = (x - 9)^2(x + 2)^2(x - 7)^3$.

- a. 12
- b. 7 ✓
- c. 8
- d. 3

The degree is the sum of the multiplicities: $2 + 2 + 3 = 7$.

34 Find the degree of $f(x) = (x + 3)^2(x - 6)^1(x - 2)^2$.

- a. 4
- b. 6
- c. 5 ✓
- d. 2

The degree is the sum of the multiplicities: $2 + 1 + 2 = 5$.

35 Find the degree of $f(x) = (x + 8)^3(x + 6)^2(x - 6)^3$.

- a. 8 ✓
- b. 9
- c. 18
- d. 3

The degree is the sum of the multiplicities: $3 + 2 + 3 = 8$.

36 Find the degree of $f(x) = (x - 3)^2(x + 8)^3(x + 1)^3$.

a. 8 ✓

b. 9

c. 3

d. 18

The degree is the sum of the multiplicities: $2 + 3 + 3 = 8$.

37 Find the degree of $f(x) = (x + 9)^2(x - 3)^1(x - 6)^2$.

a. 2

b. 4

c. 6

d. 5 ✓

The degree is the sum of the multiplicities: $2 + 1 + 2 = 5$.

38 Find the degree of $f(x) = (x + 7)^2(x - 7)^1(x - 1)^1$.

a. 4 ✓

b. 5

c. 7

d. 2

The degree is the sum of the multiplicities: $2 + 1 + 1 = 4$.

39 Find the degree of $f(x) = (x + 9)^2(x + 6)^1(x - 7)^1$.

a. 4 ✓

b. 5

c. 3

d. 2

The degree is the sum of the multiplicities: $2 + 1 + 1 = 4$.

40 Find the degree of $f(x) = (x + 7)^2(x - 1)^2(x + 6)^3$.

a. 3

b. 8

c. 7 ✓

d. 12

The degree is the sum of the multiplicities: $2 + 2 + 3 = 7$.

41 Find the degree of $f(x) = (x + 6)^1(x + 1)^3(x - 8)^2$.

a. 7

b. 3

c. 6 ✓

d. 8

The degree is the sum of the multiplicities: $1 + 3 + 2 = 6$.

42 Find the degree of $f(x) = (x - 8)^2(x + 4)^2(x + 2)^2$.

a. 2

b. 8

c. 7

d. 6 ✓

The degree is the sum of the multiplicities: $2 + 2 + 2 = 6$.

43 Find the degree of $f(x) = (x - 1)^3(x + 2)^2(x - 4)^3$.

a. 8 ✓

b. 18

c. 9

d. 3

The degree is the sum of the multiplicities: $3 + 2 + 3 = 8$.

44 Find the degree of $f(x) = (x - 9)^3(x + 1)^2(x + 3)^1$.

a. 3

b. 6 ✓

c. 7

d. 8

The degree is the sum of the multiplicities: $3 + 2 + 1 = 6$.

45 Find the degree of $f(x) = (x - 7)^3(x - 2)^3(x + 7)^3$.

a. 27

b. 3

c. 10

d. 9 ✓

The degree is the sum of the multiplicities: $3 + 3 + 3 = 9$.

Multiple choice — hard 15

46 For $f(x) = (x + 2)^1(x - 8)^3$, describe the multiplicity and graphical behavior at $x = -2$.

a. **Multiplicity 1 (odd) — the graph crosses the x-axis at $x = -2$. ✓**

b. Multiplicity 1 (even) — the graph touches and turns back without crossing the x-axis at $x = -2$.

c. Multiplicity 1 (odd) — the graph crosses the x-axis at $x = 8$.

d. Multiplicity 3 (odd) — the graph crosses the x-axis at $x = -2$.

Multiplicity 1 (odd) — the graph crosses the x-axis at $x = -2$.

47 For $f(x) = (x - 5)^4(x - 3)^2$, describe the multiplicity and graphical behavior at $x = 5$.

a. Multiplicity 2 (even) — the graph touches and turns back without crossing the x-axis at $x = 5$.

b. Multiplicity 4 (odd) — the graph crosses the x-axis at $x = 5$.

c. **Multiplicity 4 (even) — the graph touches and turns back without crossing the x-axis at $x = 5$. ✓**

d. Multiplicity 4 (even) — the graph touches and turns back without crossing the x-axis at $x = 3$.

Multiplicity 4 (even) — the graph touches and turns back without crossing the x-axis at $x = 5$.

48 For $f(x) = (x - 9)^1(x + 7)^2$, describe the multiplicity and graphical behavior at $x = 9$.

a. Multiplicity 1 (odd) — the graph crosses the x-axis at $x = -7$.

b. Multiplicity 1 (even) — the graph touches and turns back without crossing the x-axis at $x = 9$.

c. Multiplicity 2 (even) — the graph touches and turns back without crossing the x-axis at $x = 9$.

d. **Multiplicity 1 (odd) — the graph crosses the x-axis at $x = 9$. ✓**

Multiplicity 1 (odd) — the graph crosses the x-axis at $x = 9$.

49 For $f(x) = (x - 8)^1(x - 7)^2$, describe the multiplicity and graphical behavior at $x = 8$.

a. **Multiplicity 1 (odd) — the graph crosses the x-axis at $x = 8$. ✓**

b. Multiplicity 1 (odd) — the graph crosses the x-axis at $x = 7$.

c. Multiplicity 2 (even) — the graph touches and turns back without crossing the x-axis at $x = 8$.

d. Multiplicity 1 (even) — the graph touches and turns back without crossing the x-axis at $x = 8$.

Multiplicity 1 (odd) — the graph crosses the x-axis at $x = 8$.

50 For $f(x) = (x - 5)^2(x + 9)^1$, describe the multiplicity and graphical behavior at $x = 5$.

- a. Multiplicity 2 (odd) — the graph crosses the x -axis at $x = 5$.
- b. Multiplicity 1 (odd) — the graph crosses the x -axis at $x = 5$.
- c. Multiplicity 2 (even) — the graph touches and turns back without crossing the x -axis at $x = -9$.
- d. **Multiplicity 2 (even) — the graph touches and turns back without crossing the x -axis at $x = 5$. ✓**

Multiplicity 2 (even) — the graph touches and turns back without crossing the x -axis at $x = 5$.

51 How many possible rational zeros does $f(x) = x^n + \dots + 6$ (leading coefficient 1, constant term 6) have, by the Rational Zero Theorem?

- a. 4
- b. 10
- c. 6
- d. **8 ✓**

Since the leading coefficient is 1, possible rational zeros are $\pm$ (divisors of 6). 6 has 4 positive divisors, giving $2 \times 4 = 8$ possible rational zeros.

52 How many possible rational zeros does $f(x) = x^n + \dots + 10$ (leading coefficient 1, constant term 10) have, by the Rational Zero Theorem?

- a. 10
- b. 4
- c. **8 ✓**
- d. 6

Since the leading coefficient is 1, possible rational zeros are $\pm$ (divisors of 10). 10 has 4 positive divisors, giving $2 \times 4 = 8$ possible rational zeros.

- 53 How many possible rational zeros does $f(x) = x^n + \dots + 25$ (leading coefficient 1, constant term 25) have, by the Rational Zero Theorem?

a. 6 ✓

b. 8

c. 4

d. 3

Since the leading coefficient is 1, possible rational zeros are $\pm$ (divisors of 25). 25 has 3 positive divisors, giving $2 \times 3 = 6$ possible rational zeros.

- 54 How many possible rational zeros does $f(x) = x^n + \dots + 16$ (leading coefficient 1, constant term 16) have, by the Rational Zero Theorem?

a. 12

b. 10 ✓

c. 5

d. 8

Since the leading coefficient is 1, possible rational zeros are $\pm$ (divisors of 16). 16 has 5 positive divisors, giving $2 \times 5 = 10$ possible rational zeros.

- 55 How many possible rational zeros does $f(x) = x^n + \dots + 5$ (leading coefficient 1, constant term 5) have, by the Rational Zero Theorem?

a. 6

b. 2

c. 10

d. 4 ✓

Since the leading coefficient is 1, possible rational zeros are $\pm$ (divisors of 5). 5 has 2 positive divisors, giving $2 \times 2 = 4$ possible rational zeros.

- 56 A polynomial has zeros at $x = 0$, $x = 1$, $x = 2$ (each multiplicity 1) and passes through $(-1, -18)$. Find a in $f(x) = a(x)(x - 1)(x - 2)$.

a. -3

b. 3 ✓

c. 6

d. 4

$$f(-1) = a(-1)(-2)(-3) = a(-6) = -18, \text{ so } a = -18 / -6 = 3.$$

- 57 A polynomial has zeros at $x = -3$, $x = 6$, $x = -1$ (each multiplicity 1) and passes through $(0, -72)$. Find a in $f(x) = a(x + 3)(x - 6)(x + 1)$.

a. 4 ✓

b. 8

c. -4

d. 5

$$f(0) = a(3)(-6)(1) = a(-18) = -72, \text{ so } a = -72 / -18 = 4.$$

- 58 A polynomial has zeros at $x = 5$, $x = 0$, $x = 4$ (each multiplicity 1) and passes through $(-6, 660)$. Find a in $f(x) = a(x - 5)(x)(x - 4)$.

a. -1 ✓

b. -2

c. 1

d. 0

$$f(-6) = a(-11)(-6)(-10) = a(-660) = 660, \text{ so } a = 660 / -660 = -1.$$

- 59 A polynomial has zeros at $x = -6$, $x = 0$, $x = -5$ (each multiplicity 1) and passes through $(-3, 90)$. Find a in $f(x) = a(x + 6)(x)(x + 5)$.

- a. -4
- b. -10
- c. 5
- d. -5 ✓

$$f(-3) = a(3)(-3)(2) = a(-18) = 90, \text{ so } a = 90 / -18 = -5.$$

- 60 A polynomial has zeros at $x = -2$, $x = -4$, $x = 6$ (each multiplicity 1) and passes through $(0, -96)$. Find a in $f(x) = a(x + 2)(x + 4)(x - 6)$.

- a. 4
- b. 2 ✓
- c. -2
- d. 3

$$f(0) = a(2)(4)(-6) = a(-48) = -96, \text{ so } a = -96 / -48 = 2.$$